Zehua Wu

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EDUCATION

University of Pennsylvania, Philadelphia, USA — Sept. 2024 - May 2026
School of Engineering and Applied Science
Master of Science in Engineering: Electrical Engineering
GPA: 3.44/4.0

Southeast University Chengxian College, Nanjing, China — Sept.2020 - Jun. 2024
School of Electronic and Computer Engineering
Bachelor of Engineering: Automation
GPA: 3.59/4.0 | Ranking: 4/58 (Top 6%)

SKILLS

Programming Languages: Python, Java, C++, SQL, JavaScript
Frameworks: Spring Boot, FastAPI, Flutter, React, Node.js, .NET, PyTorch
Technical Infrastructures: Redis, Kafka, ChromaDB, Docker, AWS, GCP
Tools: Postman, MATLAB, Git, Codex, arm Keil5 MDK, Arduino

PROFESSIONAL EXPERIENCE

Google | Software Engineering Intern — Jun.2025 - Aug.2025
Project: Offline-First Multimodal Local Content Retrieval System
Tech Stack: Flutter, TensorFlow Lite, BERT, MobileCLIP, ChromaDB

- **Retrieval Pipeline**: Designed and implemented a modular pipeline spanning file I/O, parsing, embedding generation, vector storage, and retrieval. Supported **5** file formats, indexing of **10K+ local documents**, and incremental updates to reduce redundant parsing and embedding computations.
- **On-Device Multimodal Search**: Integrated BERT-based text embeddings and MobileCLIP image encoders using TensorFlow Lite. Implemented hybrid keyword and vector search with ChromaDB, achieving **85% Recall@10** and **P95** retrieval latency **under 300 ms**. Applied **INT8** quantization to reduce model memory footprint by **67%**.
- **Testing & Engineering**: Developed unit and end-to-end tests using Flutter Test to validate retrieval functionality and cross-platform compatibility. Participated in code reviews and cross-platform integration.

SELECTED PROJECTS

Penn Medicine·Zhi Huang Lab | Research Software Developer — Dec.2025 - Jun.2026
Project: TissueLab - AI Agentic System for Digital Pathology | Philadelphia, PA
https://tissuelab.org/
Tech Stack: Python, FastAPI, Node.js, OpenSeadragon, GPT-5, GCP, Docker, GitHub Actions

- **Agent Engineering**: Implemented runtime exception handling for pathology diagnosis workflows. Designed a unified API response contract to standardize error reporting and frontend-backend error propagation.
- **WSI Performance Optimization**: Optimized server-side Whole Slide Image (WSI) concurrent loading, achieving a **4.07x** improvement in image reading speed and reduced latency during interactive multi-resolution viewing with OpenSeadragon.
- **AI Model Integration & Evaluation**: Orchestrated pathology foundation models to automate the analysis of tumor necrosis, inflammation, and edema. Integrated human-in-the-loop correction and achieved **92.6%** diagnostic agreement with pathologist hand grading assessments.
- **DevOps & Collaboration**: Used Git, Docker, and GitHub Actions-based CI/CD workflows to support automated deployments and integration testing on GCP, contributing to SDK development, code reviews, and release quality.

OnlineOrder | Full-Stack Agent-assisted Food Ordering Application Jan.2025 - Apr.2025
Tech Stack: Java, Spring Boot, FastAPI, PostgreSQL, Redis, Kafka, Docker, JWT, GPT-4.5, WebSocket

- **High-Concurrency Checkout**: Designed an asynchronous checkout pipeline using **Redis Lua scripts for atomic inventory reservations, Kafka for request buffering, and PostgreSQL for order persistence**. Achieved **1458.82 QPS** with **128.40 ms P99 latency** and zero inventory overselling in load tests.
- **Data Consistency**: Implemented idempotent order processing with **Idempotency-Key**, PostgreSQL conditional updates, and compensating transactions to prevent duplicate orders, inventory overselling, and inconsistent Redis/database states.
- **Event-Driven Notification**: Published order lifecycle events, including payment completion, cancellation, merchant confirmation, and fulfillment, through Kafka and real-time status updates to customers and merchants via WebSocket.
- **Agent Orchestration**: Built an Agent Orchestrator that exposed Spring Boot APIs as callable tools for menu search, cart management, asynchronous checkout and order tracking. Evaluated **300** custom test scenarios, achieving **91%** tool-calling accuracy and a **90%** task success rate.
- **Session Management & Memory:** Implement Redis-backed multi-turn session management to track working states and user preferences, enabling context-aware interactions.
- **Deployment**: Containerized PostgreSQL, Redis, and Kafka using Docker Compose for reproducible deployment.

Industrial Robot Positioning Accuracy Improvement Research Nov.2023 - Jun.2024
Tech Stack: Matlab, Python, Forward Kinematics, Optimization Algorithms, Neural Networks
NSFC-Funded Research Project (**No. 51905258**) | Nanjing, China

- **Kinematics Modeling**: Developed a modified D-H kinematic model for a **Staubli TX60** industrial robot, establishing a geometric parameter identification and error compensation framework to improve end-effector positioning accuracy.
- **Experimental Calibration**: Conducted multi-pose calibration experiments using a **Leica AT960** laser tracker to collect high-precision end-effector measurements. Established a parameter identification workflow based on measured robot poses and kinematic error analysis.
- **Optimization & Error Compensation**: Utilized intelligent optimization algorithms and neural network to identify kinematic parameters and correct pose deviations, reducing end-effector position error by **90.42%** and orientation error by **80.19%**.

PUBLICATIONS

- "Research on Robot Parameter Identification Based on Black-winged Kite Optimization Algorithm," IAENG International Journal of Computer Science, 2025.
- "Compensation of Industrial Robot Positioning Accuracy Based on Multi-layer Perceptron," Instrument Technique & Sensor, 2025
- "Robot Geometry Parameter Identification Method Based on Multi-Objective Differential Evolution Algorithm," China Measurement & Test, 2024
- "A Multi-Model Detection Method and System for Intelligent Transportation," Publication No. CN116311100A, 2023

LEADERSHIP & VOLUNTEER EXPERIENCE

Institute of Dermatology, Chinese Academy of Medical Sciences Feb.2021 - Apr.2021
Volunteer | Nanjing, China

- **Team Leadership & Coordination**: Organized and coordinated a student volunteer team. Assigned volunteer responsibilities, managed on-site services, and led the team in providing patient guidance and assistance.